



Original Article

A Wearable Monitoring Device for COVID-19 Biometric Symptoms Detection

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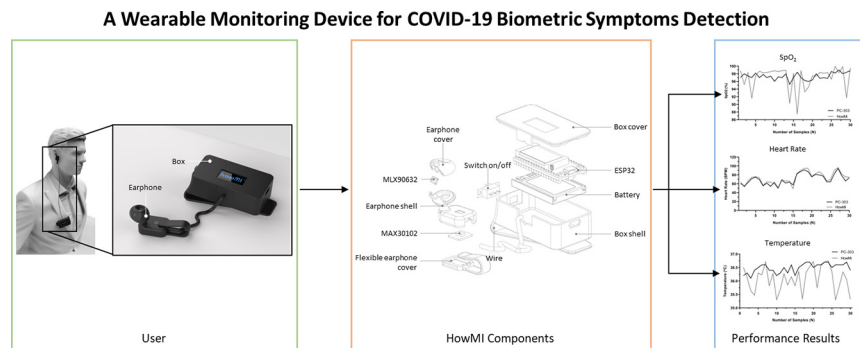
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HIGHLIGHTS

- A wearable for continuous, non-invasive COVID-19 biometrics monitoring.
- Device designed for ear, with infrared thermometer and PPG sensor.
- Reliable temperature & heart rate measurement, but limited oximetry.
- Tested on 30 participants, compared to medical-certified station.
- Wearable found to be user-friendly and comfortable.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: Monitoring COVID-19 symptoms has become a critical task in controlling the spread of the virus and preventing hospitalizations. Aiming to contribute to efficient monitoring solutions, this article presents the development and testing of a wearable device capable of continuous monitoring biometric signals associated with the presence of COVID-19, such as the heart rate, the blood oxygen saturation, and the body temperature.

Methods: To ensure continuous monitoring the device is designed to be worn in the ear. Here, the temperature is measured through a non-contact infrared temperature sensor placed inside the ear canal while the heart rate and the pulse oximetry signals are monitored through a photoplethysmography reflective sensor positioned at the earlobe.

The proposed device's performance was evaluated by comparing it against a medical certified station. Usability and ergonomics were assessed through users' questionnaires. Additionally, experiments were performed to evaluate the hearing loss when the proposed device is in use. Data was acquired from 30 individuals of different sex, aged between 20 and 43 years old. In relation to usability and ergonomics the variation in ear dimensions was accessed and related to the device's comfort limitations.

Results: The temperature measurement produced a moderate correlation ($R = 0.42$), despite a higher standard deviation was found in the proposed solution. This is due to the limited variability in temperature data, creating a short measuring range, as only healthy people were tested. The heart rate measurement also showed good correlation ($R = 0.96$), with the proposed solution showing good repeatability with a standard deviation of 6.06 BPM, however, the SpO2 measurement was suboptimal ($R = 0.14$).

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The ergonomic evaluation revealed that most participants found the device shape comfortable, but some found the dimensions not adequate.

Additionally, the device was found to be user-friendly, with most participants reporting that they found it to be intuitive, and none reported a major loss in hearing in a normal conversation, however, there's a negligible loss of approximately 0.56 dB.

Conclusions: During this study, it was possible to develop and evaluate a wearable device that was suggested for monitoring biometric signals. The device demonstrated great reliability in temperature and heart rate measurement but showed limitations in the accuracy of pulse oximetry. The main contribution of this work is the evaluation of a continuous non-invasive monitoring concept for COVID-19 related biometric signals, which indicates good applicability in the case study.

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1. Introduction

COVID-19 has caused a global crisis impacting everyone, but especially vulnerable groups like the elderly, disabled, and those with chronic diseases [1][2][3]. To prevent the spread of the virus, social distancing measures have been implemented, but they have also resulted in increased social isolation of older people, which can make it difficult to accurately monitor their physiological parameters and detect signs of illnesses [4]. These groups of people are often less capable of supporting themselves in isolation and need more assistance. Furthermore, there has been a growing trend of shifting medical care from hospitals to homes, to prevent the saturation of medical centres. This trend has created a need to remotely monitor a person's health status using devices that reduce the need for in-person interactions, but at the same time, these devices should also provide healthcare professionals with a comprehensive and real-time view of a person's health status.

Wearable electronic accessories, such as activity trackers and smartwatches, have become increasingly popular in the recent years, due to their ability to provide a metric view of users health, through continuous measurements using biometric sensors [5][6]. Unlike scheduled medical appointments, these wearable devices allow for the continuous monitoring and registering of different bio signals. This type of analysis differs from the norm, where physicians typically compare physiological values obtained during an annual testing session with the average values of the population.

Specifically, COVID-19 is associated with various physiological changes that can be monitored by biometric sensors. Metrics such as resting heart rate, heart rate variability, blood pressure, and respiratory rate can serve as evidence of an infection [7]. Elevated body temperatures and arterial oxygen saturation (SpO₂) are other clinical values that can indicate COVID-19 symptoms, due to the high prevalence of fever and respiratory changes. Early detection in the variation of these biometric parameters in the face of abnormal values may be a good indication of the presence of the disease. Allowing rapid intervention prevents the progression of the disease and hospitalizations due to exacerbations.

The surge of smartwatches and fitness trackers led to the increase of scientific investigation into wearable sensors. These daily use devices are usually designed to be used on the wrist, but also on ears and most of scientific studies focus on the viability of signal acquisition and algorithms for the processing of biometrics values such as the heart rate and the blood oxygen saturation, and do not check the design and ergonomics of the proposed solutions [8][9][10]. This aspect can be detrimental for in-ear devices that have a wearable use in mind. There is a lack of study on the development of a complete device that is ready to market or in a pilot phase. Another gap found is the number of tests performed, usually a small population sample is tested. In this sense, this paper aims to fill these gaps and present a more detailed study related to the design of a biometric ear-based wearable device.

Therefore, this study shows a proof of concept of new wearable device termed "HowMI". This wearable intends to offer end-users more confidence, supporting healthcare professionals with an innovative, more accurate information care management tool and allowing faster responses to an emergency scenario.

This paper is organized into eight sections. In section 2, the state of art with some examples of similar systems are given, as well as their limitations. In section 3, the system development steps are presented. Section 4 describes several experiments that were conducted to compare the performance of the proposed system with a medical certified monitoring station that has been approved by the FDA (Food and Drug Association). In Section 5, the results of tests are presented. In section 6, an interpretation and discussion of the obtained results are given. Additionally, some future suggestions to improve the proposed system are also given. Section 7 describes the study limitations and finally, in section 8 the conclusions are drawn.

2. State of art

This section is organized into two subsections. The first one focuses on the analysis of comparable devices that use similar sensory technologies. In the second one, relevant design features are extracted and indicated.

2.1. Biometric signals and wearable devices

Wearable electronic devices, like activity trackers and smartwatches, provide a metric vision of personal health, through embedded biosensors [5][6]. This allows continuous monitoring of personal health, avoiding the need for routine check-ups in medical facilities, which are normally scheduled after some symptom is felt.

COVID-19 prevention can be done with various types of sensors from contactless technologies like thermal cameras [11] to simple body thermometers. The person heart-rate, arterial blood oxygen saturation and breathing rate are also affected by this virus [12].

Heart rate signals are often used to detect viral diseases such as COVID-19 [13]. These illnesses frequently cause physiological alterations in the human body, manifesting usually in an increased frequency of heartbeats. In viral infections, the modification of the abovementioned bio signals can occur hours or days before the first symptoms. This represents the response of the immune system to the infection [14].

Electrocardiogram (ECG) and photoplethysmography (PPG) are the most used methodologies for non-intrusive sensing. ECG technology analyses the electric activity of the heart, usually from epidermic adhesives. PPG technology uses light to measure the difference in blood volume flowing through the blood vessels [15]. This technology is more suitable for continuous monitoring because the sensor can be placed in various positions on the body, for example the wrist, earlobe, finger and chest. Because of that,

they are usually integrated on smartwatches, fitness trackers, and smart earphones.

As COVID-19 affects the human respiratory function, it is fundamental to analyse its respiration to prevent more degenerative effects. For that, it is common to use blood oxygen saturation (SpO₂) that can be extracted from a PPG sensor.

Studies for the acquisition of biometric signals for monitoring the heartbeat and/or oxygenation of the blood in the earlobe and the warning of pathologies began from an early age. In 1974, Stern [16] presented a study on the existence of very little change in pulse volume in the ear lobe as a function of the usual types of psychological stimuli, being so the earlobe an inadequate place to record vasomotor reactivity (VRM) - defined as the percent change in blood flow velocity in response to changes in CO₂ - helps identifying individuals who are at risk for or have evidence of low flow ischemia.

Vogel et al. [9] proposed a 24/7 optical monitoring system in which the central component is a micro-optical reflection sensor placed inside the auditory canal. Through the result of photoplethysmography curves (wrist photo-plethysmography PPG), the metrics of heart rate, oxygen saturation and respiratory rate were estimated. This device was proposed to determine early cardiovascular risk factors to take preventive measures. However, only during the last years, further advances were found for these monitoring technologies, resulting in the mass manufacture of commercial sensors to obtain the metrics under study. Lueken et al. [10] presented an ear system for the development of physiological and psychological stress based on photoplethysmography, acceleration, and temperature measurements. The system consisted of a MAX30100 (Maxim Integrated, San Jose, United States) [17] reflective sensor, a temperature sensor to measure inside the ear, and an accelerometer to detect movements. The goal was to monitor individuals who were exposed to psychologically demanding tasks.

Differently, Bu et al. [18] developed a system for measuring blood pressure (BP). However, a wrist or arm bracelet over a long period of time has a significant impact on daily activities. The main objective was to reduce this insomuch and to perform continuous monitoring of blood pressure for the diagnosis and treatment of diseases.

Davies et al. [8] developed a wearable device to estimate oxygen saturation in the blood. The authors reported that the non-invasive estimation of blood oxygen saturation by pulse oximetry was of vital clinical importance, from the discovery of sleep apnoea to the recent ambulatory monitoring of hypoxemia in the late post-infective phase of COVID-19. They also added suggestions to mitigate the problem of the smaller amplitude of the photoplethysmogram of the auditory canal in relation to the finger and the fact that continuous monitoring in the ear is more convenient since it is less intrusive in clinical settings and daily life, when compared to conventional finger measurement.

Correia et al. [19] presented a study to validate the use of a PPG sensor in the earlobe to extract heart rate variability data from stress-inducing tasks. The authors successfully validated the use of a commercial Bluetooth PPG sensor - named "James One" - compared to industry-leading devices. The PPG device tended to overestimate certain HRV (Heart rate Variability) metrics.

Body temperature was the most used method to prevent COVID-19 transmission as fever usually represents a virus infection [20]. However, high body temperature appears in an advanced state of the infection being so not suitable as a standalone measurement to prevent COVID-19 transmission [21]. It is known that there are several places to perform body temperature measurements, and this varies from place to place. For example, the mean tympanic temperature is between 35.76-37.61 °C, the rectal is between 36.32-37.76 °C and the axillary between 35.01-36.93 °C.

Moreover, body temperature varies with the age group and gender of the person [22].

The method of measuring body temperature through central thermometers, for example in the restorative, is more accurate, but it is extremely invasive [23]. There are studies comparing body temperature by digital rectal thermometer and infrared tympanum thermometer where it was depicted a good agreement between the two methods, proving that both methods are interchangeable. Here, the researchers also mentioned that patients prefer the use of the tympanum thermometer because it was not so invasive and because it took less time to acquire the temperature [24].

Seshadri et al. [25] and Channa et al. [7] analyzed the wearable devices market. These devices are in its majority wrist-worn and only very few were used on other body parts. The OURA ring, for example, can monitor the parameters previously described and is being studied for COVID-19 monitoring.

Most studies test the applicability of sports wearables as monitoring devices, failing to correctly evaluate its feasibility for clinical purposes, mainly in older people. Moreover, sometimes the devices are too complex and not user friendly for a public that commonly lacks technical know-how. These devices are also very expensive because they have additional features, that are not required for patient monitoring.

2.2. Design

Design has become a very important discipline in the conceptual process. According to Bedworth et al. [26], it has been demonstrated that the concept formulation stage determines 70% of the production cost of a product. The design of a part aimed at humans should always be studied, taking into consideration ergonomics and anthropometrics, as it allows design to be adjusted to suit specific markets.

Specifically, for this work, ear anthropometry contains studies on the size and shape of the outer ear, as an anthropometric reference for the design of ear-related products. The outer ear, also referred to as the auricle or pinna, includes helix, antihelix, trapezoid, anti-tragus, lobule, cymba concha, cavum concha, triangular fossa, intertragic incisure and ear canal (Lee et al.) [27].

There are numerous articles where different studies are conducted to analyse the dimensions and shapes of the ear in order to understand which dimensions follow for product design. Therefore, some articles who studied ear dimensions and were important for the development of the device presented in this paper were studied (Table 1).

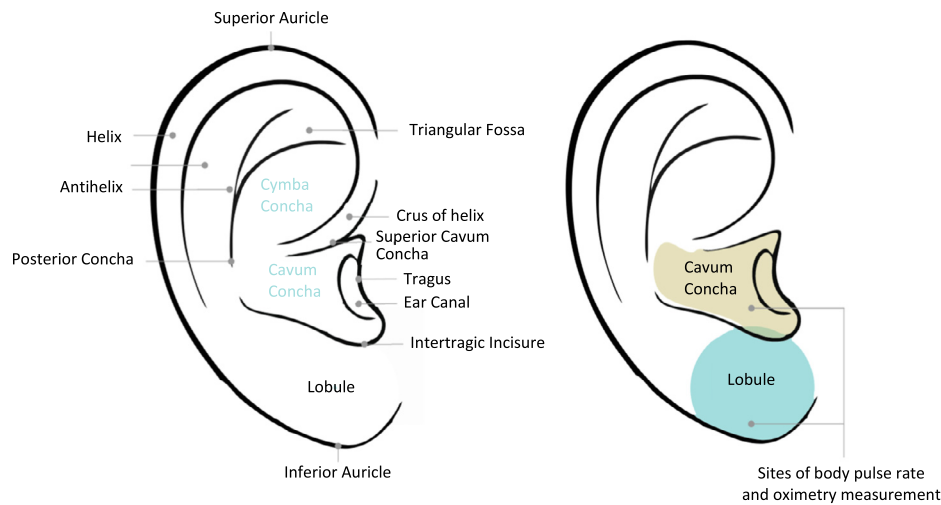
Ear related products, such as headphones, earphones, and different forms of hearing aids, are widely used in our daily lives. These products are designed to fit different regions of the outer ear. Today there is a wide range of products, of different shapes, colours, and sizes. Focusing on the objective of the study, namely the measurement of temperature, oximetry, and heart rate in the ear, a study of ear dimensions and usability was performed. The most relevant dimensions of the study are the height from the intertragic incisure to the superior cavum concha, the length from the tragus to the posterior concha, and the total ear height, as illustrated in Fig. 1.

Most devices that take blood oxygen or heart rate readings on the earlobe consist of a spring-loaded piece for attaching the device to the lobe or the finger, respectively. The Nonin 8000Q2 Reusable Ear Clip Pulse Oximetry Sensor (Nonin Medical, Plymouth, USA) [31] and the ES-3212-9 (EnviteC by Honeywell, North Carolina, USA) [32] are examples of earlobe pulse oximetry reading devices. One of the devices on the market that reads all three metrics is the Cosinuss One (Cosinuss GmbH, Munich, Germany) [33]. This device measures all three metrics inside the ear canal, in addition to movement. To measure the temperature, a contact

Table 1

Studies on ear dimensions.

Authors	Type of study	Study population	Ear height (mm)	Cavum concha height (mm)	Cavum concha width (mm)
Lee et al. [28]	3D scan	100 right ears of Koreans people aged between 20 and 59 years old	63.50	17.200	16.900
Chiou et al. [29]	Silicone mould	220 ears of Taiwanese people aged between 20 and 40 years old	-	19.995	16.995
Lee et al. [27]	3D ear scan images	230 Koreans	63.30	16.950	16.850
		96 Caucasians	60.40	14.850	16.400
Ban et al. [30]	3D scan anthropometry	186 Koreans	58.86	15.870	17.360
		124 Caucasians	57.65	15.210	16.660

**Fig. 1.** Ear measurements.

thermometer based on resistance is used. This type of technology takes longer to read the body temperature than a contactless infrared sensor. However, measuring temperature inside the ear canal provides a more accurate measurement, compared to external body locations.

3. Methods – HowMI wearable device

The development of the wearable prototype is composed of two main parts, namely, the system architecture and design. The description of the architecture includes the hardware used and the software developed for data processing and transmission. The design depicts the evolution of the wearable prototype, having in consideration the size and geometry restrictions of ear measurements. The proposed device, named “HowMI”, comprises two significant components: one of the parts is similar to an earphone placed in the ear and incorporates the biometric sensors as well as some electronic signal processing; the other part integrates the microcontroller with Bluetooth communication protocol and the battery. Fig. 2 shows the proposed system implementation as well as all the components of the enclosure.

3.1. System architecture (hardware and software)

3.1.1. Hardware

The HowMI device utilizes several sensors to obtain biometric information including heart rate, blood oxygen saturation, and body temperature. These non-intrusive sensors aim to provide an accurate estimate of the patient’s condition. The device architecture is displayed Fig. 3. Later, the different parts that make up the hardware will be detailed.

- 1) Pulse Oximeter:** Photoplethysmography is a technique used to obtain information about heart rate (HR) and peripheral oxygen saturation (SpO₂). These signals were obtained with the front end sensor MAX30102 (Maxim Integrated, San Jose, USA) [34]. This sensor implements the reflective type of PPG chip. This sensor already integrates internally LED’s, photodetectors, optical components, as well as ambient light cancellation. The MAX30102 contains a red LED (RED) and an infrared LED (IR) with wavelengths of 660 nm and 875 nm, respectively, and has reduced dimensions, 5.6 mm × 3.3 mm × 1.55 mm. The analogue-to-digital converter (ADC) has several resolutions and different sampling rates. In this study a pulse width of 411 μs and 200 samples per second were the best options. The MAX30102 is presented in wearables, fitness assistant devices, smartphones, and tablets.
- 2) Temperature:** There are two distinct techniques for acquiring body temperature. The first technique is based on contact thermometers, that is, there is a resistance that changes according to the temperature variation of the human. In this method, body temperature is easily manipulated since the skin temperature differs from the human core temperature. Another approach is to use non-contact infrared sensors placed on the surface of the eardrum, because the eardrum is highly perfused by hot arterial blood. In this work, MLX90632 (Melexis, Tessenderlo, Belgium) [35] was integrated into the wearable to obtain the variation of human body temperature. The MLX90632 is an IC (Integrated Circuit) miniature SMD (Surface Mounted Device) thermometer for accurate measurement of contactless temperature, especially in thermally dynamic environments and when available space is limited. The sensor used is medically certified and has an accuracy of ±0.2 °C in the human body temperature range. Its dimensions are 3 mm × 3 mm × 1 mm.

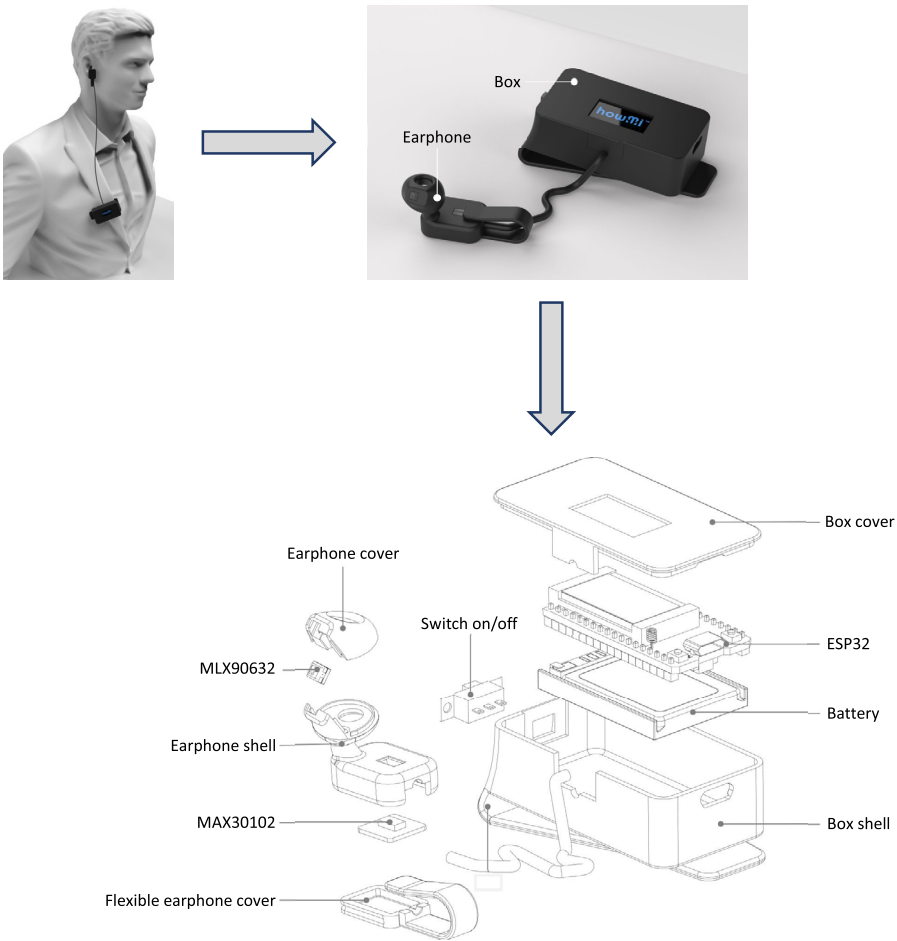


Fig. 2. System overview.

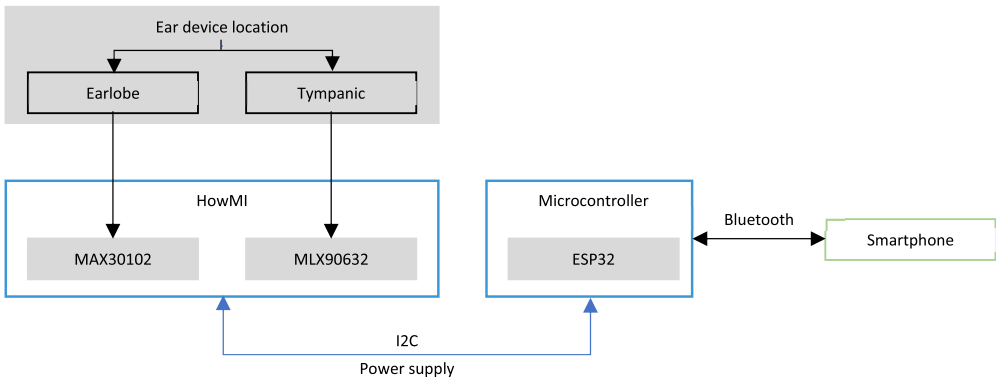


Fig. 3. HowMI overview.

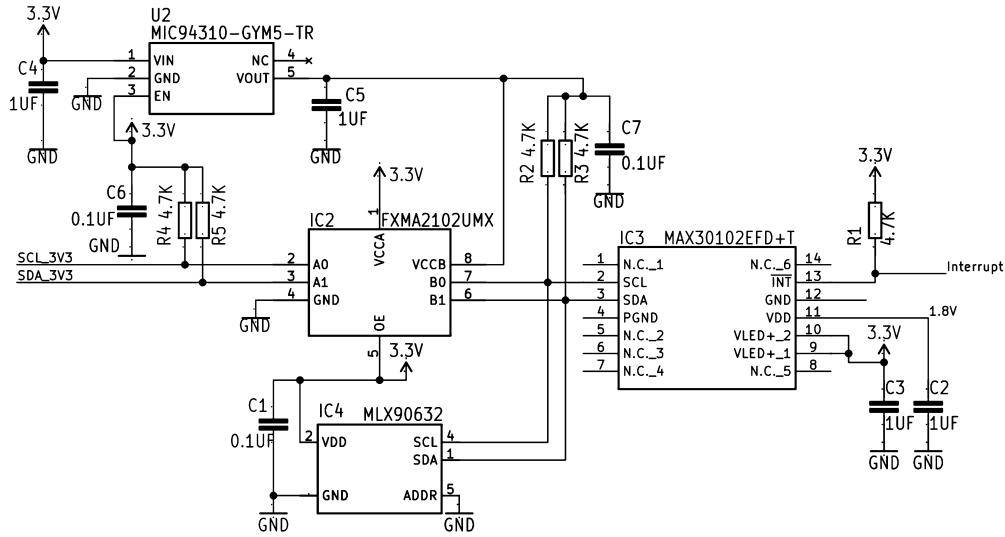


Fig. 4. Circuit schematic of the earpiece part of the HowMI device.

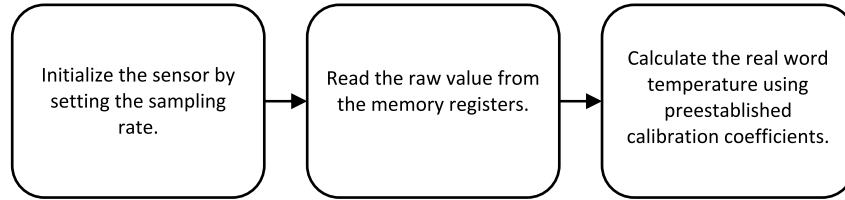


Fig. 5. Flowchart with the basic steps of temperature calculation.

3) **Microcontroller:** To process the data acquired by the sensors and subsequent wireless transmission a ESP32 (Espressif Systems, Shanghai, China) microcontroller was used. The choice for this microcontroller was because it has Bluetooth already integrated which facilitates later with the connection to a smartphone application as well as there are various development boards with integrated circuits for battery management, for example, HELTECs “Wi-Fi Kit 32” [36]. In addition, this board has already incorporated a display to visualize the biological data after the acquisition. To power the entire system, a 550 mAh lithium battery measuring 40 mm × 30 mm × 4.6 mm was used.

The circuit, which incorporates both the MAX30102 and the MLX90632 sensors, decoupling capacitors, and level shifter for power and communication with the microcontroller at 1.8 V, was enclosed in the earpiece part of the system as shown in the schematic presented in Fig. 4.

3.1.2. Data acquisition

The temperature reading from the MLX90632 can be acquired directly from the memory registers of the IC as it is already calibrated for body temperature, and only post-calibration is needed.

Fig. 5 describes the basic operation steps of the MLX90632.

Reflective pulse-oximetry use LEDs to illuminate the skin tissue and a photodiode to detect the reflected signal. This reflected signal contains the light that was modulated by the volumetric changes of the arteries and capillaries in the tissue. In the calculus of SpO₂, it is assumed that only oxygenated haemoglobin and deoxygenated haemoglobin are the significant factors. PPG signals have a DC (Direct Current) component and an AC (Alternating Current) component. The non-pulsatile tissue absorbs the light and causes the DC components. Differently, the AC component is attributed to the pulsatile nature of the arterial blood.

The instantaneous heart rate can be detected by measuring the time between consecutive systolic peaks.

The heart rate measurement is only extracted from the infrared signal. The following steps describe the basic calculation of the instantaneous heart rate (Fig. 6).

The SpO₂ value is calculated from both RED and IR waveforms according to the provided algorithm from the manufacturer. First, the DC and AC components are extracted from both signals, then we find the normalized ratio of the RED wave with the IR wave. This ratio of metrics given by equation (1) is possible because the proportion of oxygenated haemoglobin in the blood affects the ratio of absorbance of infrared to red light within the PPG sensor.

$$R = \frac{AC_{red}/DC_{red}}{AC_{ir}/DC_{ir}} \quad (1)$$

Where, AC_{red} is the Alternating Current component of the red signal, DC_{red} is the Direct Current component of the red signal, AC_{ir} is the Alternating Current component of the infrared signal, and DC_{ir} is the Direct Current component of the infrared signal.

The ratio can then be used to empirically determine the SpO₂ value using a lookup table or a curvilinear approximation. This is provided by the manufacturer. The SpO₂ is calculated based on the manufacturer calibration given in equation (2).

$$SpO_2 = -45.060 \times R^2 + 30.354 \times R + 94.84 \quad (2)$$

The basic steps to calculate the SpO₂ are implemented in the library provided by the manufacturer as presented in Fig. 7.

3.1.3. Bluetooth Low Energy (BLE) protocol

To simplify wireless integration of the device, we opted to use a set of standard Bluetooth Low Energy (BLE) protocols in addition to custom services for data transmission, allowing a faster integration of the central unit. Those were tested with a generic BLE scanning

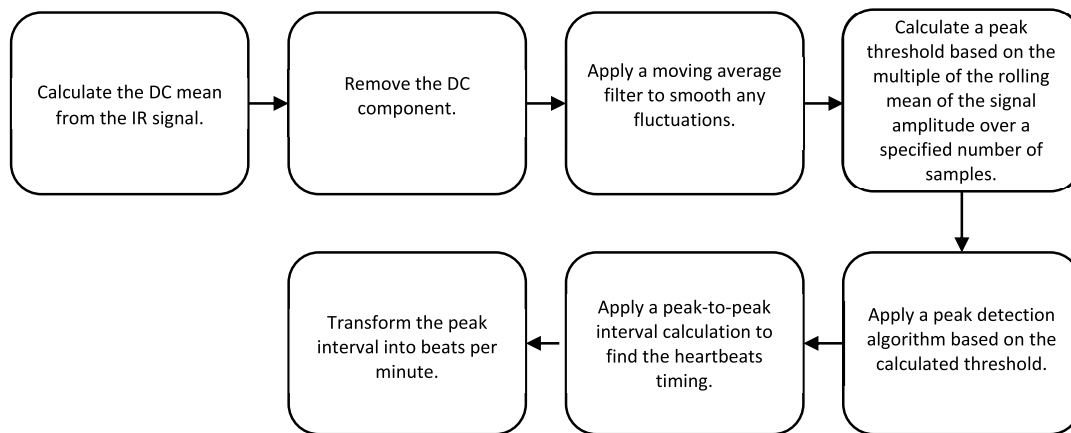


Fig. 6. Flowchart with the basic steps of heart rate calculation.

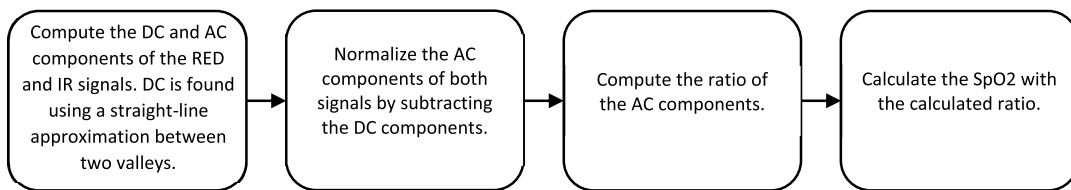


Fig. 7. Flowchart with the basic steps of SpO2 calculation.

app. The basic application functionality for continuous monitoring is described in Fig. 8.

The standard GATT (Generic Attribute) services used to transfer biometric data to the Bluetooth central were Health Thermometer (HT) and Heart Rate Profile (HR). A custom service was also implemented to provide basic configuration parameters, related to the device function and pulse oximetry.

3.2. Design

Our study aims to measure temperature, oximetry, and pulse rate. It was important to understand where the three measurements could be taken, ideally all together, so that the device would be as small dimension as possible.

Oximetry values can be obtained through a translucent part of the patient's body, such as the fingertip or earlobe. Body temperature can be measured in five locations: armpits, mouth, anus, ear, and forehead [22]. Through this first analysis we realized that there is one common site – the ear. Indeed, one of the goals of this study is to evaluate the accuracy of all measurements performed at the ear.

To study the shape of the device it was important to analyse the ear, understand its geometry and how the sensors would need to be positioned to take the correct readings. Although all ears are different and there are numerous characteristics at each person, the device should be usable by most people.

According to the best locations to measure, the positions of the technology were defined: the ear canal for the MLX90632 sensor and the earlobe for the MAX30102 sensor, Fig. 9.

With definition of the locations to be measured, it was important to identify the dimensions needed for the projection of the device. The most pertinent dimensions of the study are the total ear height (D1), the concha cava length (D2) and the concha cava height (D3), as illustrated in Fig. 10.

The 3D modelling of the device was done with the CAD software SolidWorks (Solidworks, Dassault Systems, SolidWorks Cor-

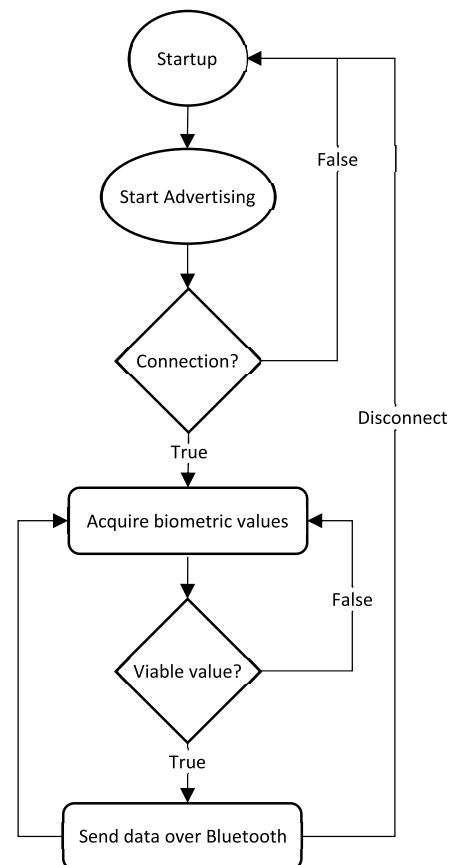


Fig. 8. Firmware overview.

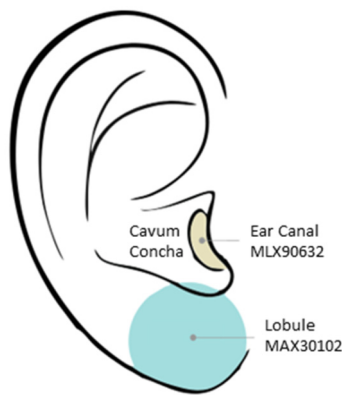


Fig. 9. Locations of the technology in the ear.

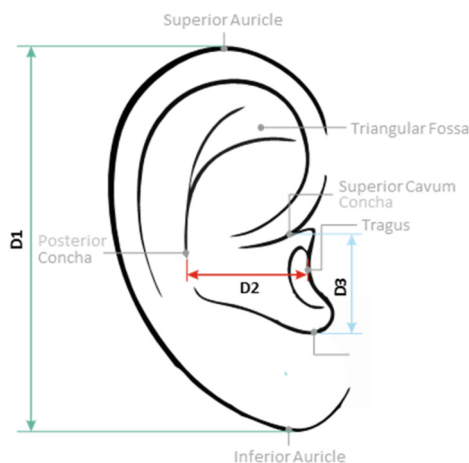


Fig. 10. Dimensions relevant to the study.

poration, Waltham, Massachusetts, United States), according to ergonomic and functional requirements. The device consists of two main parts:

- the earphone, where it is possible to measure the temperature in the ear canal and the oximetry in the earlobe; the spring allows pressure to be exerted between the earlobe and the sensor, to obtain reliable values.
- the box, which contains the device's communication technology (Fig. 2).

The device was made using additive manufacturing with an SLA (Stereolithography) 3D printer, for this we used a Anycubic Photon Mono X (Shenzhen Anycubic Technology Co. Ltd). A mixture of standard 3D print resin and a flexible resin was implemented to give the device a more comfortable fixture. A ratio of 70% hard resin (FLGPGR02, Formlabs) to 30% flexible resin (FLFLGR01, Formlabs), made the best mixture for this application, reducing the brittleness of the hard resin and allowing the frame to slightly conform to the user ears.

After printing the parts were cleaned and cured in the Anycubic Wash & Cure Plus Machine (Shenzhen Anycubic Technology Co. Ltd). Fig. 11 shows how the proposed solution should be placed in the user's ear.

4. Experiments

This section details the three experiments performed to evaluate the performance of the proposed wearable device. The first experiment is related to ergonomics and usability, assessing fac-

tors related to ease of use and comfort. In the scope of comfort, the loss of hearing caused using the device was also evaluated.

In the second and third experiments, the temperature, heart rate and SpO2 metrics were evaluated, respectively.

All the tests further described were performed on a group of 30 participants in order to allow a relative higher confidence. This test group considered Caucasian people with a balanced ratio of gender. From the research team knowledge the participants did not disclose any condition that could be detrimental to the tests as, for example, hearing constraints or particular cardiac conditions.

4.1. Ergonomics and usability

Ergonomic and usability tests were conducted to identify any potential issues and possible areas for improvement after the device was three-dimensionally modelled and printed in resin.

The primary goals of the experiment were to evaluate the ease of placement and comfort of the device in the ear, the effect of the flexible lid in reading values, and the appropriate positioning of the sensors. The experiment also aimed to assess the ease of clipping the box to the clothes, and if the weight and size of the box caused any discomfort. Additionally, it was crucial to observe the first impressions of the test participants to check if they understood how the device should be used without prior knowledge.

The users in this experiment used the device for a period between 10 and 20 min. Following the experiment and their analysis, certain aspects required improvement. The feedback from users was instrumental in adjusting the device's design.

4.2. Hearing loss

The test of the effect of the proposed device in the hearing of the participant was performed to assess their ability to hear sounds at different frequencies, with and without the HowMI placed in the ear.

The tests were performed in a sound isolated room, during this time sine waves with different frequencies were generated and their amplitude was progressively increased. The participants were instructed to indicate when they started hearing those distinctive sounds. The frequency chosen were those typical of an audiogram: 250, 500, 1000, 2000, 4000, 8000 Hz. To measure the volume of the generated sound waves with high fidelity a "SC160 Sound level Meter" (CESVA instruments, Barcelona, Spain) [37] was used.

These tests were performed with and without the device, to access the quantitative effect of the device in the hearing of the participants. Additionally, the participants were asked to give their feedback about the restrictions of the HowMI device during a normal conversation.

4.3. Temperature

Performance tests were conducted to evaluate the performance of the proposed wearable device by comparing its biometric readings with those provided by a portable health monitoring system. The system used for comparison was the "PC-303 All-in-One Health Monitor" (Shenzhen Creative Industry Co., Ltd., Shenzhen, China) [38]. The results were analysed and compared to understand what improvements were necessary.

The temperature measurement was conducted in a single instance, in which the temperature read of the HowMI solution and the PC-303 was annotated. The participants were instructed to place the developed device on one ear and wait for the sensor to provide the temperature reading. After this, the temperature was acquired by the monitoring station. These tests were performed at a different room temperature, to reduce the impact of the ambient temperature on the values measured.

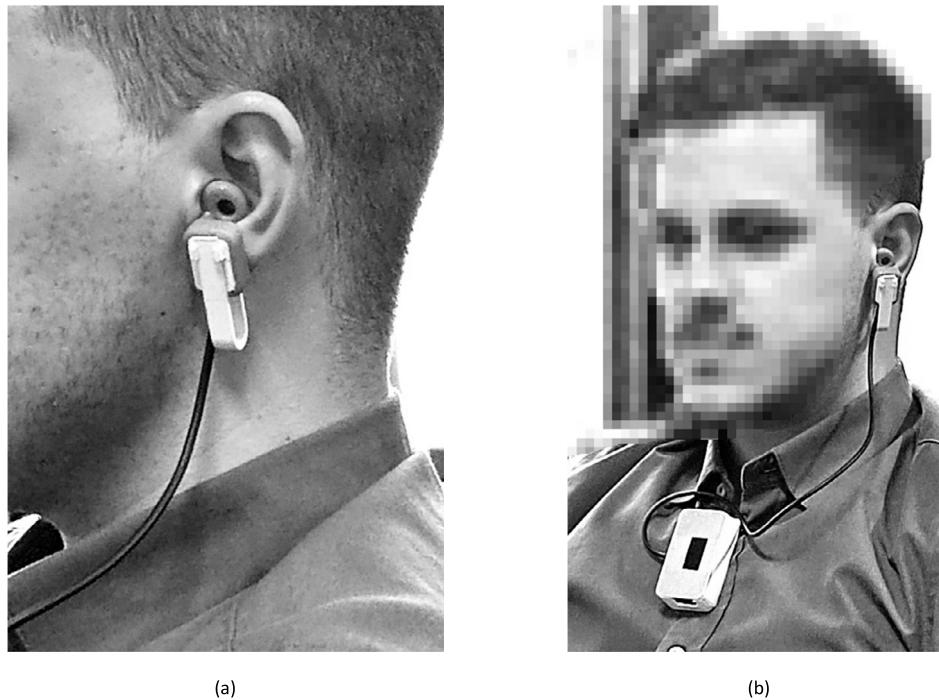


Fig. 11. Designed use of the HowMI device in a test environment. (a) Lateral view; (b) Front view.

4.4. Heart rate and SpO2

The heart rate and blood oxygen metrics were tested in two stages. Initially, the PPG sensor on the proposed device was validated for everyone by placing the volunteer's index finger over the sensor while simultaneously using the PC-303 pulse oximeter probe on an adjacent finger to acquire readings from both devices. To ensure accurate readings, participants were instructed to remain as still as possible for the duration of the test. A time frame of 5 min was defined for each participant, during which the first five relevant readings were registered. These values were updated at 15 s intervals. If less than three values were recorded the test was considered invalid and the previous results were discarded.

The objective of the second test was to evaluate the performance of the device on the defined position. The participants had the HowMI device placed on the ear while the PC-303 pulse oximeter probe was placed on the index finger. Like the previous tests, the data was acquired for a period of 5 min, and only the first five valid measurements were recorded.

The average of these multiple measurements is a more accurate representation of the participant's real health condition.

5. Results

The following section presents the results obtained from the tests conducted on the proposed device, including the evaluation of ergonomics, temperature acquisition, heart rate and blood oxygen metrics.

5.1. Ergonomics and usability

The results of the measurements taken from the test subjects provided valuable insights for the development of the prototype. The device was dimensioned based on the average dimensions of the ear "cavum concha". The test subjects were divided into two groups based on their feedback regarding the device's fit: those who found it uncomfortable and prone to falling (G1) and those who found it comfortable (G2). The participants in G1 have an average horizontal interior measurement (D2) of 16.80 mm, while

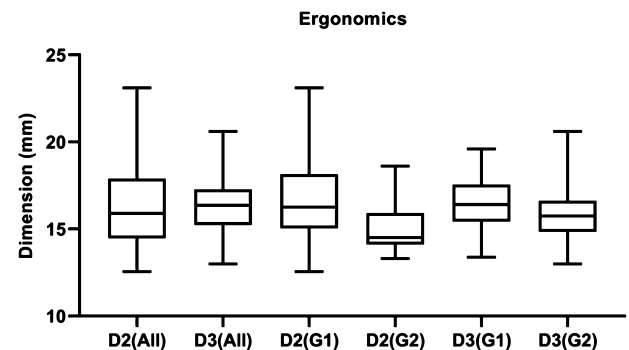


Fig. 12. - Measurements distribution.

for participants in G2 the same measurement was 15.15 mm, as shown in Table 2. The vertical measurement (D3) was similar for both groups, with an average of 16.58 mm for G1 and 16.03 mm for G2.

It is worth noting that G1 has a higher range of width and height compared to G2. This can be observed in Fig. 12 where the distribution of G1 is shown to be greater than that of G2.

Table 3 highlights some improvements that were identified based on the user's review. Other commentaries are a downside to the use of wearables in the ear, such as the conflict with the use of jewellery, such as earrings, or hearing aid devices. Also due to the material used in the 3D printing skin irritation is prone to appear after prolonged use.

The tests were helpful in determining that the device performed as intended, but it was evident that some ergonomic improvements were needed to enhance its usability and overall product quality.

5.2. Hearing loss

Concerning the experiments realized to measure the hearing loss, the differences registered are found in both Table 4 and

Table 2

Ear measurements (mm).

	Dimensions (mm)					
	D2 (All)	D3 (All)	D2 (G1)	D2 (G2)	D3 (G1)	D3 (G2)
Number of Samples	30	30	21	9	21	9
Minimum	12.56	12.99	12.56	13.30	13.38	12.99
Maximum	23.10	20.60	23.10	18.60	19.59	20.60
Range	10.54	7.61	10.54	5.30	6.21	7.61
Mean	16.21	16.62	16.80	15.15	16.58	16.03
Std. Deviation	2.33	1.79	2.45	1.58	1.69	2.07

Table 3

User reviews and improvement approaches.

Reviews	Improvements
The wire placement should be adjusted to ease the placement of flexible clip.	Position the wire off-centre from the earpiece and have it protrude from the side of the casing.
After a few uses, the resin material went into plastic deformation and eventually ruptured.	Prototyping the lid in flexible PLA (Polylactide) instead of flexible resin.
A more flexible wire between the two components should be used, to not limit head movements.	Changing wire material and length.
Sensation that the earpiece could fall off.	Having a selection of multiple sizes for the earpiece.

Table 4

Average difference of decibels in the hearing of the test frequencies with and without the HowMI device.

Frequency (Hz)	125	250	500	1000	2000	4000	8000
Average Difference (dB)	0.53	0.29	0.40	0.34	0.68	0.41	1.27

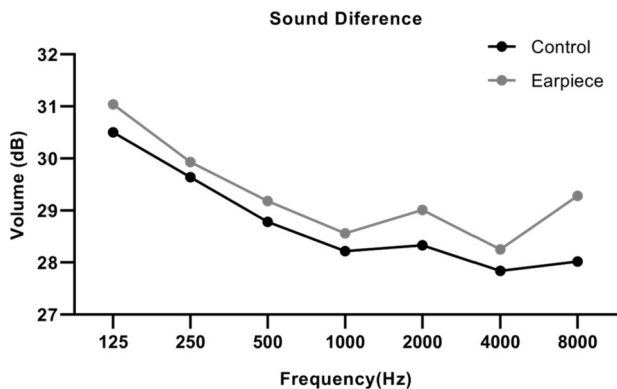
**Fig. 13.** Evaluation of the hearing loss - HowMI device in use.

Fig. 13. Overall, the use of HowMI do not appear to be associated with a high hearing loss, being the average difference of 0.56 dB.

5.3. Performance results

5.3.1. Temperature

The values obtained from eligible tests, presented in Table 5, show that the performance of this pilot device is acceptable, with an average difference of 0.40 °C when compared to the portable station. The medical station, predictably, show a higher quality. From the graphs shown in Fig. 14, one understands the similarities of the measurement of the HowMI solution and the biometric station.

Comparing all temperature values, we observed a correlation coefficient (r) of 0.42, Table 6. This suboptimal result is due to the limited variability in temperature data, creating a short measuring range, as only healthy people were evaluated. This statistical analysis was performed in Graphpad Prism 8 (GraphPad Software, San Diego, CA, USA).

Table 5

Temperature performance (°C).

	Temperature (°C)	
	PC-303	HowMI
Number of samples	30	30
Minimum	36.10	35.30
Maximum	36.70	36.75
Range	0.60	1.45
Mean	36.49	36.08
Std. Deviation	0.17	0.46

Table 6

Temperature statistical hypothesis.

HowMI vs PC-303	
Pearson R	
Pearson r	0.42
95% confidence interval	0.06 to 0.67
R^2	0.17

5.3.2. Heart rate & SpO2

The analysis of the results involved averaging the values obtained from each individual and conducting a statistical analysis to reduce the impact of out-of-the-ordinary values. This approach provided a more representative picture of the person's health status.

The evaluation of the heart rate and blood oxygen metrics provided insightful results. Specifically, the HowMI device exhibited an average standard deviation of 4.66 BPM (Beats per Minute) for heart rate measurement on the fingertip, which is higher than that of the PC-303 at 2.34 BPM, Table 7. In the earlobe the average standard deviation of the HowMI solution is 6.06 BPM, and the station is similar to the previous at 3.39 BPM. These values show high repeatability in the measurement of heart rate for both systems.

Concerning the SpO2 measurements, the HowMI system exhibited an average standard deviation of 0.59% when measured on the

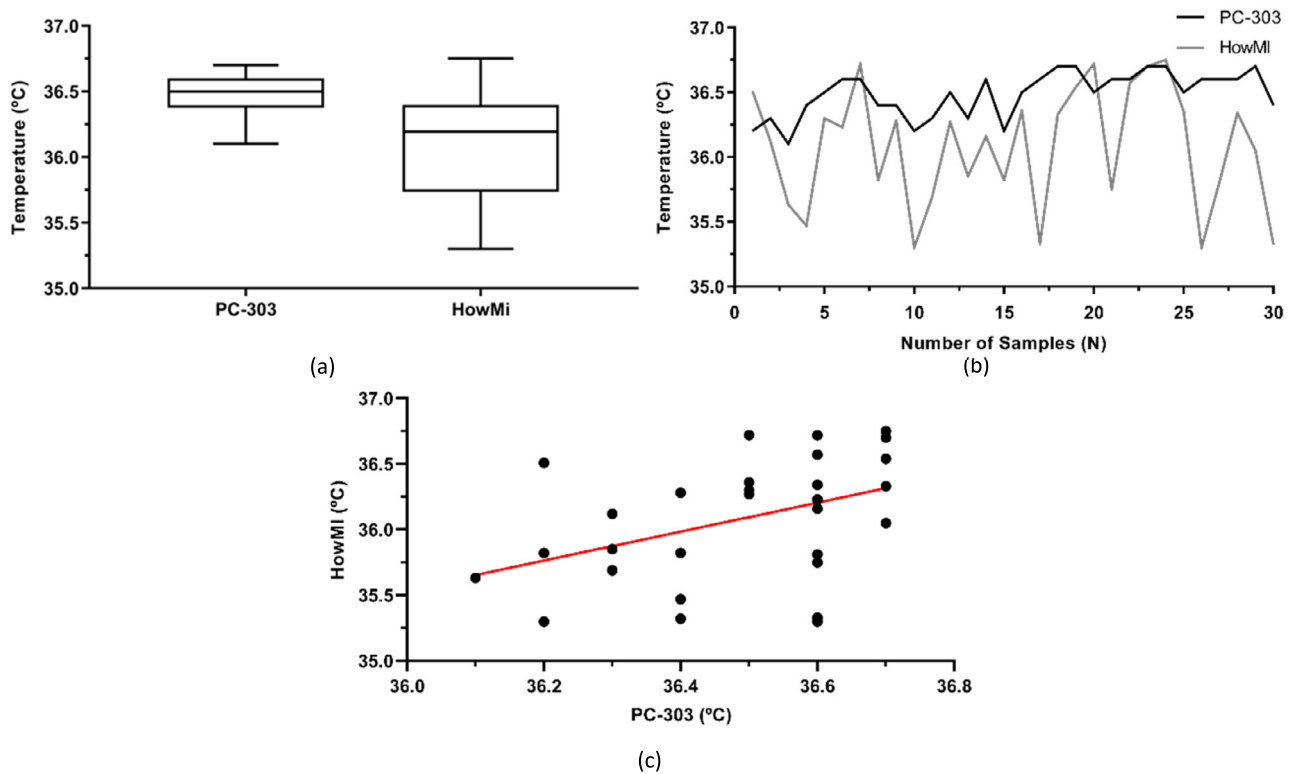


Fig. 14. Analyse of temperature data. (a) Temperature distribution; (b) Temperature Measurement; (c) Correlation of temperature.

Table 7

Descriptive statistics of Heart Rate (HR) for the mean and standard deviation value (BPM) of each participant.

	Heart Rate (BPM)							
	HowMI (Finger)		PC-303(Finger)		HowMI (Earlobe)		PC-303(Finger)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Minimum	45.40	0.00	44.60	0.71	49.00	2.19	50.40	0.55
Maximum	104.8	13.75	97.20	5.55	96.60	18.20	94.40	11.31
Range	59.40	13.75	52.60	4.84	47.60	16.01	44.00	10.77
Mean	72.58	4.66	70.00	2.34	73.21	6.06	70.47	3.39
Std. Deviation	13.30	3.65	13.12	1.26	13.19	3.48	11.83	2.58

Table 8

Descriptive statistics of Pulse Oximetry (SpO2) for the mean and standard deviation value (%) of each participant.

	Pulse Oximeter (%)							
	HowMI (Finger)		PC-303(Finger)		HowMI (Earlobe)		PC-303(Finger)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Minimum	97.32	0.04	96.00	0.00	87.35	0.02	95.20	0.00
Maximum	99.71	2.91	98.80	3.83	99.94	9.14	99.00	1.10
Range	2.39	2.87	2.80	3.83	12.44	9.12	3.80	1.10
Mean	98.52	0.59	97.57	0.45	96.95	1.80	97.36	0.31
Std. Deviation	0.64	0.68	0.70	0.71	3.10	2.41	0.91	0.33

fingertip and 1.80% on the earlobe. The monitoring station reading showed a better repeatability at 0.45% and 0.31%, respectively, as presented in Table 8.

The average values of each individual were then compared, and the resulting correlation coefficients were analysed for both stations. The heart rate metric showed a high correlation coefficient of 0.96, while the SpO2 metric only a low correlation coefficient of 0.14 (as shown in Table 9). This result is due to the limited variability in SpO2 data, creating a short measuring range, as only healthy people were tested.

Fig. 15(a) shows a graph comparing the average heart rate for each individual with both probes on the finger, while Fig. 15(c)

presents a comparison of the proposed solution when measured on the earlobe. The correlation between the HowMI system and the station is positive, with the difference between the measurements being insignificant when observed on an extended scale of 0 to 100 BPM. Fig. 15(b)(d) shows the graphs with the average blood oxygen saturation level for everyone tested, and the difference in accuracy of the two systems can be seen. On the fingers both systems give responses, within a normal physiological range, they are all above 95% as expected. However, in the lobe reading the HowMI solution shows some readings outside the physiological acceptable value. Fig. 15(e)(g) shows the good correlation between both systems in

Table 9
HR & SpO2 statistical hypothesis.

	Earlobe		Finger	
	HR	SpO2	HR	SpO2
Pearson r	0.96	0.14	0.95	0.24
95% confidence interval	0.93 to 0.98	−0.23 to 0.47	0.90 to 0.98	−0.14 to 0.55
R ²	0.93	0.02	0.91	0.06

the reading of heart rate. Fig. 15(f)(h) represents the correlation is the SpO2 measurement.

6. Discussion

This study presents a possible continuous monitoring solution able for early detection of COVID-19. It is also reinforced with various test studies in a large population sample compared to other similar articles, one first on ergonomics and usability and a second on a comparison between the wearable and a medical approved system.

The ear dimensions gathered from multiple people, and the answers from the usability questionnaire, allowed us to observe a tendency for the device to fall when the width of the earpiece was less than or equivalent to the interior diameter of the ear concha. The vertical measurement was similar in both groups. Therefore, the values are not considered critical to the device ergonomics. The main problems reported on the questionnaire, were the fact that the headset was large and at the time caused discomfort and the connecting wire between the headphones and the power module was short, restricting movements and leaving the hearing device.

The temperature comparison proved that the developed solution is reliable at reading the temperature from a person in the ear channel. Our device tends to read a lower value than the reference system with an average difference of approximately 0.4 °C. This represents an acceptable performance in temperature measurement; however, this value could be improved, a better isolation of the ear channel with encapsulation made with softer materials could prove a viable solution, and with better guidelines, the position that a person places the earpiece could improve. During testing, there was one participant whose measurements did not correlate with the expected physiological range interval values. This outlier was subsequently removed from further testing being later verified that was caused by the incorrect positioning of the device. This occurrence shows that there is a probability that user errors can affect the measured values and so a clear instruction on their use needs to be given.

Regarding the PPG sensor performance test, the first results showed that the pilot device is working as expected, however, the results show slightly different values, for heart rate which uses only the infrared light the results are a good indication of the usability of the device in the finger. This test on the finger showed that the proposed solution tends to overshoot the measured value. The SpO2 values however are not as revealing as the red-light signal is more susceptible to external noise.

In the earlobe, the beats per minute of the heart rate, are also very similar to the monitoring station. The SpO2 values acquired in the earlobe are also less than the desired. The significant performance difference is attributed to the effects of the ambient light noise from external light sources, the difficulty of removing motion artefacts, incorrect placement of the sensing probe on the earlobe, and as vascular activity is inferior on the earlobe. The results show that a more substantial study of the optomechanical integration of the sensor in the earlobe is required.

During usability tests and evaluation of the measuring algorithm the research team faced two problems, the first is related to the wire connecting the earpiece to the microcontroller and the

second with the spring that helps to secure the wearable to the earlobe. Since the wire that connects the wearable to the microcontroller is centred on the device, when placing it on the ear and fixing it with the spring, it ends up hindering its placement. In relation to the spring that attaches to the wearable because it is a mixture of flexible resin with hard, it ends up deforming and losing the initial characteristics. This material turns out not to be the most suitable since it would force the user to constantly change this component. These two problems will be solved in the next versions of the prototype.

Although a usable device was created, it has some limitations. One of the limitations of the wearable is the material. The device was printed on 3D printing resin, and it is therefore a somewhat uncomfortable material for continued use. One solution would be to replace it for thermoplastic polyurethane (TPU), as this material is already widely used in wearables wrist straps and medical accessories [39]. This material is biocompatible and more comfortable. Another limitation of the device is related to the diversity of oximetry values acquired in the earlobe in certain people. This last point is related to the correct fit between the wearable shape and the different geometries of the ear, making extremely hard to develop a universal device. Therefore, it is envisioned to create different adapters or different device sizes to guarantee optimal fit between device and ear. Nevertheless, further research on this point is mandatory to optimize the current solution.

Although not investigated in this study, one critical aspect that needs to be studied and optimized in future research is the energy consumption of the device. The device's operational duration depends on the transmission rate of the biometric signal over Bluetooth, which is currently set to 30 min intervals during normal operation. Based on empirical calculations, the current hardware iteration of the device has a battery life of about 15 h, as shown in Appendix A. These results suggest that the current iteration of the device is suitable for daily use and can be charged during nighttime. However, an improvement in power consumption could be achieved by refining the hardware and optimizing the firmware, including the sleeping modes of the sensors.

As a final comment, this study focused on the quality of signal acquisition in junction with the design of the product. In this way, we cannot conclude that a change in the measured biometric signal can automatically and autonomously predict a COVID-19 infection with high fidelity, as an improvement in the performance of the device is needed. One future landmark of this study is the implementation of a function to predict the likelihood of the user having COVID-19 or other illness, based on the biometric data provided. This must be done with the help of healthcare professionals, as to not create false positives indications. By incorporating machine learning algorithms and statistical models, it may be possible to identify patterns in the biometric data that are indicative of COVID-19 infection. With further research and development, such a function could potentially be integrated into the device and provide users with an additional tool to help monitor their health and prevent the spread of the virus. Another improvement would be the addition of extra sensory features like movement tracking and ambient sensors. A refinement of the hardware is also necessary to reduce the dimensions of the various parts that make up the proposed wearable. Considering the biometric signals measured and

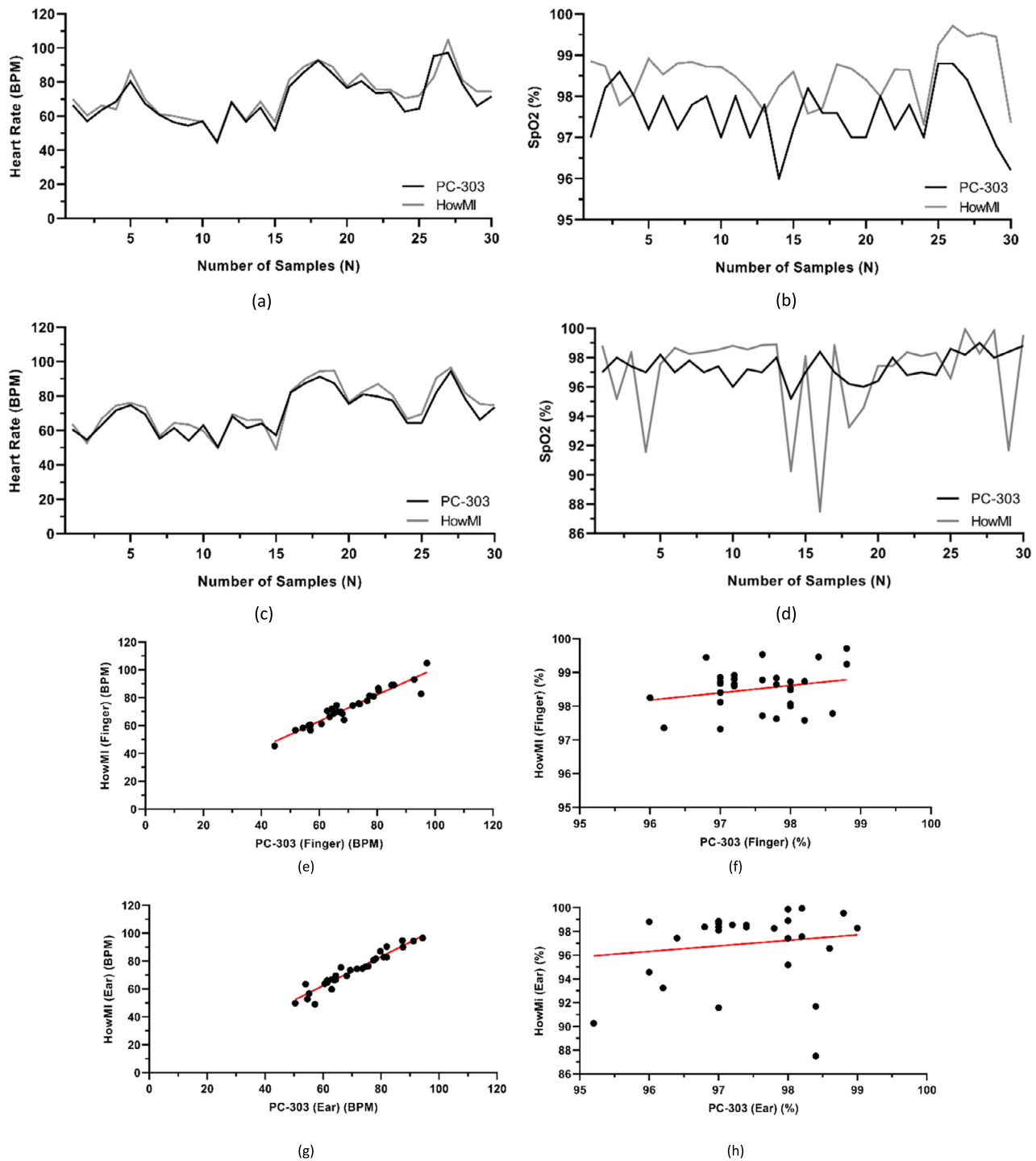


Fig. 15. Analysing the heart rate and SpO2 data. (a) Heart rate finger; (b) SpO2 finger; (c) Heart rate earlobe; (d) SpO2 earlobe; (e) Correlation of heart rate in finger; (f) Correlation of SpO2 in finger; (g) Correlation of heart rate in earlobe; (h) Correlation of SpO2 in earlobe.

analysed it is possible to assume that the proposed device could be applied to other industries or scenarios where the prevalence of hypoxic situations is of high probability (mining, mountaineering, among others).

7. Study limitations

Despite the promising results, there are some limitations within this study. One major limitation is the short measuring range of the tests made, as only healthy people were available during the pandemic. This resulted in a low correlation coefficient between some measured values and the true values, particularly for oxygen saturation. Another limitation is the lack of diversity, the study population was limited to a specific demographic. Additionally, the study focuses on quality of signal acquisition and did not consider other factors that may affect the accuracy of the biometric measurements like ambient noise and movement, there's also a lack of long-term follow-up as this study was limited to short-term measurements.

The results shown, in particular, those related to the ergonomics and usability of the device, that they are reliant of subjective opinions, and therefore may induce some bias.

Despite these limitations, the study provides valuable insights into the challenges of acquiring accurate biometric measurements and highlights the need for further research and development in this area.

8. Conclusion

This study aimed to develop a wearable device capable of detecting early signs of COVID-19 by measuring human biometric signals of temperature, heart rate and SpO₂. According to the obtained results the proposed system was successfully implemented and validated. Our study demonstrated a feasible solution for acquiring biomedical signals in the ears, although good signal acquisition can be challenging.

Informed consent and patient details

The authors declare that this report does not contain any personal information that could lead to the identification of the patient(s).

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Table A.1

Battery life considering transmission rate.

Transmission Rate (s)	30	60	300	600	900	1200
Battery Life (≈h)	4	5	10	12	13	14

Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship.

Declaration of competing interest

The authors declare that they have no known competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

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Appendix A

Although not explored in this paper, some critical expects to study and optimize in a future development are related with the energy consumption. The operation duration of the device is based on the transmission rate of the biometric signals over Bluetooth Low Energy (BLE) protocols. In normal operation the device is planned to transmit data over BLE at 30 min intervals. Based on an empirical calculation, and in its current hardware iteration, this results in a battery life of about 15 h. To reach this value, we divided the battery capacity by the average device current consumption over time. Considering a worst-case scenario, where a higher transmission rate is necessary to track the health condition of a critical patient, the battery life of the device would be drastically reduced, Table A.1.

These values were obtained considering that the current consumption of the entire device is 242.6 mA, when in transmission mode, and 33.6 mA, when the transmission module of the ESP32 is in sleep mode. 30 s were considered the most pertinent time for transmission, having in account the possibility that the device fails to connect to the central station and needs to try to reconnect. The charging time of the device was not measured, however, considering an established charging current of 200 mA, a theoretical charging time close to 3 h is possible. These values show that the current iteration of the device is relevant for daily use and that it can be charged during night-time. Nevertheless, an improvement of power consumption could be performed with a refinement of the hardware, and optimization of the firmware - considering the sleep modes of the sensors.

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